

Commerce XI Maths App

Unit 2 Theory (Pages 20-30)

Detailed answers with explanations and steps

Total questions: 41 | +2 correct | -1 wrong

Question 1 - MCQ · PG-20 · TH-Q1

A function f from set A to set B associates each element x in A with how many elements y in B ?

Options

- A. Exactly one
- B. At least two
- C. No element
- D. Any number, including none

Correct answer: Exactly one

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is Exactly one.

Question 2 - MCQ · PG-20 · TH-Q2

If $f: A \rightarrow B$ is a function, set A is called the

Options

- A. domain of f
- B. range of f
- C. co-domain only
- D. image of f

Correct answer: domain of f

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is domain of f .

Question 3 - MCQ · PG-20 · TH-Q3

If $f: A \rightarrow B$ and $y = f(x)$, then y is called the

Options

- A. image of x under f
- B. pre-image of x
- C. domain of f
- D. inverse of x

Correct answer: image of x under f

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is image of x under f .

Question 4 - MCQ · PG-20 · TH-Q4

If $f: A \rightarrow B$ and $y = f(x)$, then x is called the

Options

- A. pre-image of y under f
- B. image of y
- C. co-domain of y
- D. range of y

Correct answer: pre-image of y under f

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is pre-image of y under f .

Question 5 - MCQ · PG-20 · TH-Q5

A relation from A to B is not a function if an element of A

Options

- A. has no image or has more than one image in B
- B. has exactly one image in B
- C. has the same name as an element of B
- D. is joined by a curved arrow

Correct answer: has no image or has more than one image in B

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is has no image or has more than one image in B .

Question 6 - MCQ · PG-20–21 · TH-Q6

A function $f: A \rightarrow B$ is one-one if

Options

- A. distinct elements in A have distinct images in B
- B. every element of B has two pre-images
- C. all elements of A have the same image
- D. A and B contain the same elements

Correct answer: distinct elements in A have distinct images in B

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is distinct elements in A have distinct images in B.

Question 7 - MCQ · PG-21 · TH-Q7

A function $f: A \rightarrow B$ is onto if

Options

- A. every element of B is an image of some element of A
- B. every element of A has two images
- C. some element of B has no pre-image
- D. $f(A)$ is a proper subset of B

Correct answer: every element of B is an image of some element of A

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is every element of B is an image of some element of A.

Question 8 - MCQ · PG-21 · TH-Q8

For an onto function $f: A \rightarrow B$, which equality holds?

Options

- A. $f(A) = B$
- B. $f(A) = A$
- C. $f(A) = \emptyset$
- D. $A \cap B = \emptyset$

Correct answer: $f(A) = B$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $f(A) = B$.

Question 9 - MCQ · PG-21 · TH-Q9

A one-one and onto function is also called

Options

- A. bijective
- B. constant
- C. into

D. step

Correct answer: bijective

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is bijective.

Question 10 - MCQ · PG-21 · TH-Q10

A function is into when

Options

- A. at least one element of the co-domain has no pre-image
- B. every element of the co-domain has a pre-image
- C. every input has two outputs
- D. its range equals its co-domain

Correct answer: at least one element of the co-domain has no pre-image

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is at least one element of the co-domain has no pre-image.

Question 11 - MCQ · PG-21–22 · TH-Q11

In the ordered-pair representation of a function, the domain is the set of

Options

- A. first components
- B. second components
- C. all components
- D. products of components

Correct answer: first components

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is first components.

Question 12 - MCQ · PG-21–22 · TH-Q12

In the ordered-pair representation of a function, the range is the set of

Options

- A. second components
- B. first components

- C. ordered pairs themselves
- D. unused co-domain elements

Correct answer: second components

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is second components.

Question 13 - MCQ · PG-22 · TH-Q13

For $y = f(x) = 2x + 1$, $f(x)$ is read as

Options

- A. f of x
- B. f multiplied by x
- C. inverse of x
- D. x of f

Correct answer: f of x

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is f of x .

Question 14 - MCQ · PG-22 · TH-Q14

In graphical form, the domain is read from the extent of the graph on the

Options

- A. x-axis
- B. y-axis
- C. line $y = x$
- D. origin only

Correct answer: x-axis

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is x-axis.

Question 15 - MCQ · PG-22 · TH-Q15

In graphical form, the range is read from the extent of the graph on the

Options

- A. y-axis

- B. x-axis
- C. line $y = x$
- D. origin only

Correct answer: y-axis

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is y-axis.

Question 16 - Enter answer · PG-22 · TH-Q16

For $f(x) = 2x^2 - 3x + 4$, find $f(7)$.

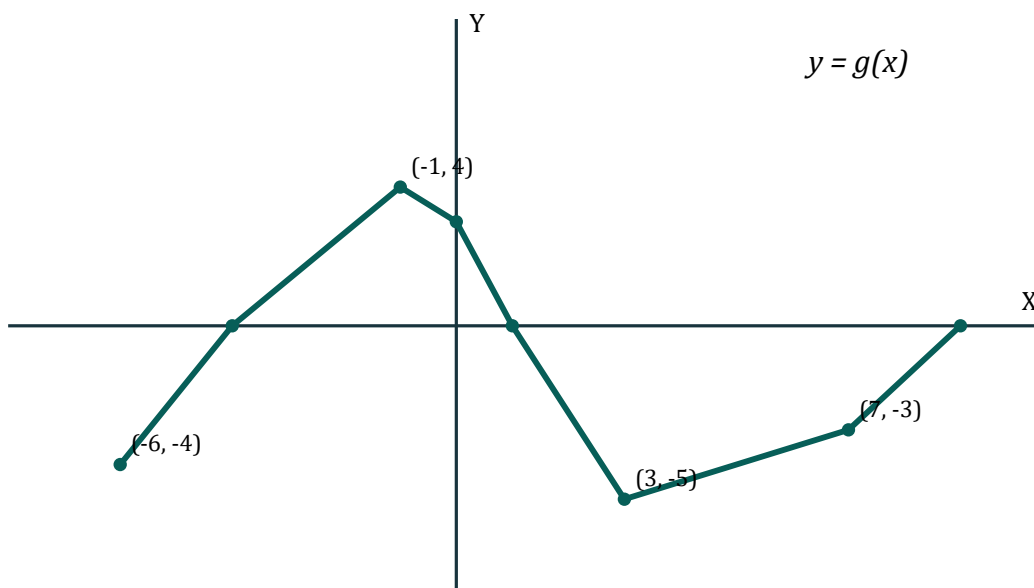
Correct answer: 81

Explanation and steps

Apply the stated definition, substitute the given value, or simplify the required expression. This gives 81. The other choices use an incorrect substitution, operation, domain restriction, or function order.

Question 17 - MCQ · PG-22 · TH-Q17

Using the graph of $y = g(x)$ shown below, $g(-4)$ equals



Options

- A. 0
- B. -4
- C. 3
- D. -5

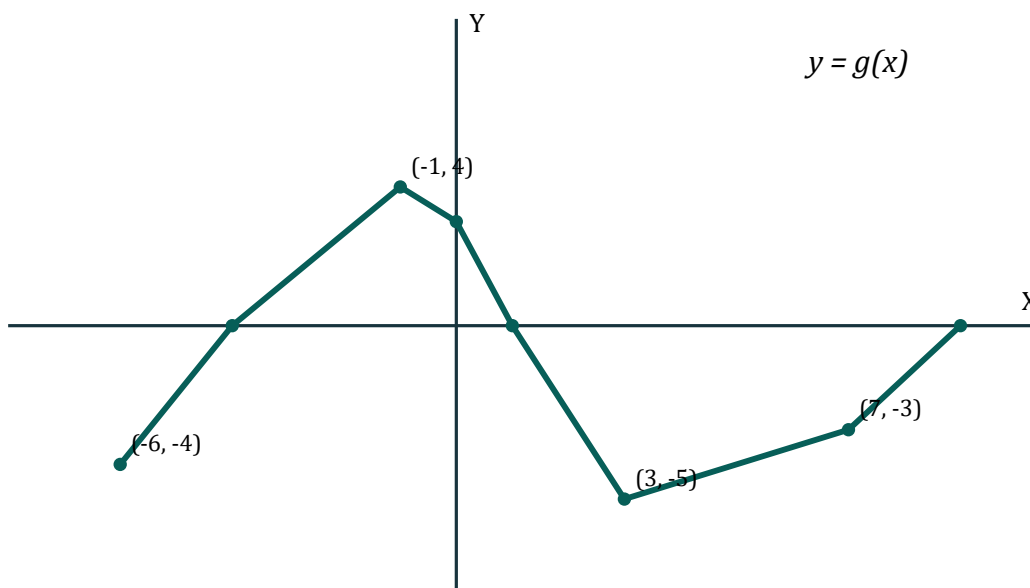
Correct answer: 0

Explanation and steps

Apply the stated definition, substitute the given value, or simplify the required expression. This gives 0. The other choices use an incorrect substitution, operation, domain restriction, or function order.

Question 18 - MCQ · PG-22 · TH-Q18

Using the graph of $y = g(x)$ shown below, $g(3)$ equals



Options

- A. -5
- B. 0
- C. 3
- D. 5

Correct answer: -5

Explanation and steps

Apply the stated definition, substitute the given value, or simplify the required expression. This gives -5. The other choices use an incorrect substitution, operation, domain restriction, or function order.

Question 19 - MCQ · PG-22–23 · TH-Q19

If $f(x) = 3x^2 - x$ and $f(m) = 4$, then m equals

Options

- A. $4/3$ or -1
- B. 1 or $-4/3$

- C. 4 or -1
D. $1/3$ or -4

Correct answer: $4/3$ or -1

Explanation and steps

Apply the stated definition, substitute the given value, or simplify the required expression. This gives $4/3$ or -1 . The other choices use an incorrect substitution, operation, domain restriction, or function order.

Question 20 - MCQ · PG-23 · TH-Q20

A constant function $f: \mathbb{R} \rightarrow \mathbb{R}$ has the form

Options

- A. $f(x) = k$
B. $f(x) = x$
C. $f(x) = ax + b$, $a \neq 0$
D. $f(x) = 1/x$

Correct answer: $f(x) = k$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $f(x) = k$.

Question 21 - MCQ · PG-23 · TH-Q21

For the constant function $f(x) = 3$, its range is

Options

- A. $\{3\}$
B. $\mathbb{R}$
C. $[0, \infty)$
D. $\mathbb{R} - \{0\}$

Correct answer: $\{3\}$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $\{3\}$.

Question 22 - MCQ · PG-24 · TH-Q22

The identity function on $\mathbb{R}$ is defined by

Options

- A. $f(x) = x$

B. $f(x) = 1$

C. $f(x) = x^2$

D. $f(x) = |x|$

Correct answer: $f(x) = x$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $f(x) = x$.

Question 23 - MCQ · PG-24 · TH-Q23

The graph of the linear function $f(x) = ax + b$ has slope and y-intercept respectively

Options

A. a and b

B. b and a

C. $-b/a$ and a

D. a and $-b/a$

Correct answer: a and b

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is a and b.

Question 24 - MCQ · PG-24 · TH-Q24

The graph of $f(x) = ax + b$ is increasing when

Options

A. $a > 0$

B. $a < 0$

C. $a = 0$ only

D. $b < 0$

Correct answer: $a > 0$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $a > 0$.

Question 25 - MCQ · PG-24 · TH-Q25

For a quadratic function whose graph has minimum value k, the range is

Options

A. $[k, \infty)$

- B. $(-\infty, k]$
- C. $\mathbb{R} - \{k\}$
- D. $\{k\}$

Correct answer: $[k, \infty)$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $[k, \infty)$.

Question 26 - MCQ · PG-24–25 · TH-Q26

The domain and range of the square function $f(x) = x^2$ are

Options

- A. Domain $\mathbb{R}$, Range $[0, \infty)$
- B. Domain $[0, \infty)$, Range $\mathbb{R}$
- C. Domain $\mathbb{R}$, Range $\mathbb{R}$
- D. Domain $\mathbb{R} - \{0\}$, Range $\mathbb{R} - \{0\}$

Correct answer: Domain $\mathbb{R}$, Range $[0, \infty)$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is Domain $\mathbb{R}$, Range $[0, \infty)$.

Question 27 - MCQ · PG-25 · TH-Q27

The domain and range of the cube function $f(x) = x^3$ are

Options

- A. Domain $\mathbb{R}$, Range $\mathbb{R}$
- B. Domain $\mathbb{R}$, Range $[0, \infty)$
- C. Domain $\mathbb{R} - \{0\}$, Range $\mathbb{R}$
- D. Domain $[0, \infty)$, Range $\mathbb{R}$

Correct answer: Domain $\mathbb{R}$, Range $\mathbb{R}$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is Domain $\mathbb{R}$, Range $\mathbb{R}$.

Question 28 - MCQ · PG-25 · TH-Q28

A rational function has the form

Options

A. $f(x) = p(x)/q(x)$, $q(x) \neq 0$

B. $f(x) = ax + b$ only

C. $f(x) = ax^n$ only

D. $f(x) = \log_a x$ only

Correct answer: $f(x) = p(x)/q(x)$, $q(x) \neq 0$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $f(x) = p(x)/q(x)$, $q(x) \neq 0$.

Question 29 - MCQ · PG-25 · TH-Q29

For $f(x) = 1/x$, the domain and range are

Options

A. $\mathbb{R} - \{0\}$ and $\mathbb{R} - \{0\}$

B. $\mathbb{R}$ and $\mathbb{R}$

C. $[0, \infty)$ and $\mathbb{R}$

D. $\mathbb{R}$ and $\{0\}$

Correct answer: $\mathbb{R} - \{0\}$ and $\mathbb{R} - \{0\}$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $\mathbb{R} - \{0\}$ and $\mathbb{R} - \{0\}$.

Question 30 - MCQ · PG-26 · TH-Q30

For an exponential function $f(x) = a^x$, the required conditions on a are

Options

A. $a > 0$ and $a \neq 1$

B. $a < 0$

C. $a = 1$

D. $a = 0$

Correct answer: $a > 0$ and $a \neq 1$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $a > 0$ and $a \neq 1$.

Question 31 - MCQ · PG-26 · TH-Q31

The domain and range of $f(x) = a^x$ are

Options

- A. $\mathbb{R}$ and $(0, \infty)$
- B. $(0, \infty)$ and $\mathbb{R}$
- C. $\mathbb{R}$ and $\mathbb{R}$
- D. $\mathbb{R} - \{0\}$ and $\mathbb{R} - \{0\}$

Correct answer: $\mathbb{R}$ and $(0, \infty)$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $\mathbb{R}$ and $(0, \infty)$.

Question 32 - MCQ · PG-26 · TH-Q32

The logarithmic statement $y = \log_a x$ is equivalent to

Options

- A. $a^y = x$
- B. $x^y = a$
- C. $a^x = y$
- D. $y^a = x$

Correct answer: $a^y = x$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $a^y = x$.

Question 33 - MCQ · PG-26 · TH-Q33

The domain of the logarithmic function $f(x) = \log_a x$ is

Options

- A. $(0, \infty)$
- B. $\mathbb{R}$
- C. $[0, \infty)$
- D. $\mathbb{R} - \{0\}$

Correct answer: $(0, \infty)$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $(0, \infty)$.

Question 34 - MCQ · PG-27 · TH-Q34

For functions f and g , $(f + g)(x)$ equals

Options

A. $f(x) + g(x)$

B. $f(g(x))$

C. $f(x)g(x)$

D. $f(x)/g(x)$

Correct answer: $f(x) + g(x)$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $f(x) + g(x)$.

Question 35 - MCQ · PG-27–29 · TH-Q35

If $f: A \rightarrow B$ and $g: B \rightarrow C$, then $(g \circ f)(x)$ equals

Options

A. $g[f(x)]$

B. $f[g(x)]$

C. $g(x) + f(x)$

D. $g(x)f(x)$

Correct answer: $g[f(x)]$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $g[f(x)]$.

Question 36 - MCQ · PG-27 · TH-Q36

For functions f and g , $(f/g)(x)$ is defined as $f(x)/g(x)$, provided

Options

A. $g(x) \neq 0$

B. $f(x) = 0$

C. $g(x) = 0$

D. $f(x) = g(x)$

Correct answer: $g(x) \neq 0$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $g(x) \neq 0$.

Question 37 - MCQ · PG-28 · TH-Q37

If f is one-one and onto and $f(x) = y$, then

Options

- A. $f^{-1}(y) = x$
- B. $f^{-1}(x) = y$ always
- C. $f(y) = x$ always
- D. f^{-1} does not exist

Correct answer: $f^{-1}(y) = x$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $f^{-1}(y) = x$.

Question 38 - MCQ · PG-28 · TH-Q38

If f and g are inverse functions, then

Options

- A. $f[g(x)] = x$ and $g[f(x)] = x$
- B. $f[g(x)] = 0$
- C. $f(x) + g(x) = x$
- D. $f(x) = g(x)$ for every x

Correct answer: $f[g(x)] = x$ and $g[f(x)] = x$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $f[g(x)] = x$ and $g[f(x)] = x$.

Question 39 - MCQ · PG-29–30 · TH-Q39

The range of the signum function $\text{sgn}(x)$ is

Options

- A. $\{-1, 0, 1\}$
- B. $\mathbb{R}$
- C. $[0, \infty)$
- D. Set of integers

Correct answer: $\{-1, 0, 1\}$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $\{-1, 0, 1\}$.

Question 40 - MCQ · PG-30 · TH-Q40

For the absolute value function $f(x) = |x|$, the range is

Options

- A. $[0, \infty)$
- B. $\mathbb{R}$
- C. $(-\infty, 0]$
- D. $\{-1, 0, 1\}$

Correct answer: $[0, \infty)$

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $[0, \infty)$.

Question 41 - MCQ · PG-30 · TH-Q41

For the greatest integer function $f(x) = [x]$, the range is

Options

- A. $\mathbb{I}$, the set of integers
- B. $\mathbb{R}$
- C. $[0, \infty)$
- D. $\{-1, 0, 1\}$

Correct answer: $\mathbb{I}$, the set of integers

Explanation and steps

This follows directly from the definition or result stated in Unit 2. Therefore the correct completion is $\mathbb{I}$, the set of integers.